

Statistics Notes: Diagnostics, ANOVA, and Multiple Comparisons

Lecture Notes

Diagnostics ANOVA

Diagnostics

We start with the linear model $Y = X\beta + \epsilon$. We must check assumptions.

- **Normality:** qq plot
- **Mean Zero Assumption:** Plot residuals against fitted values & check if there is a pattern.
- **Equality of (σ^2) across levels :**
 - Plot $\hat{\epsilon}_i^2$ against fitted values.
 - **Levene's Test:**
 - For categorical X : Is $\hat{\epsilon}_i^2$ higher for certain categories of X ?
 - Test if the mean of $\hat{\epsilon}_i^2$ for each category is the same. This can be done using **ANOVA**.
- **Normality Assumption:** (e.g., Q-Q plot of residuals)

One-Way ANOVA

Consider a model with one factor at multiple levels.

- **Model:** $Y_{ij} = \mu_i + \epsilon_{ij}$
 - $i = 1, \dots, I$ (groups)
 - $j = 1, \dots, n_i$ (observations per group)
- **Hypothesis Test:** The F-test for ANOVA tests $H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_I$.
- **Problem:** Suppose we run the one-way ANOVA and the test for equality of μ_i 's (or α_i 's) rejects. This implies at least one α_i is different.
- **Question:** Which ones are different? This leads to **pairwise comparisons**.

$$H_0 : \alpha_a = \alpha_b$$

$$a, b \in \{1, \dots, k\}, a \neq b$$

Any two levels of the factor (a, b).

$$\frac{\hat{\alpha}_a - \hat{\alpha}_b}{s.e.(\hat{\alpha}_a - \hat{\alpha}_b)} \sim t_{N-I}$$

$$s^2 \perp \hat{\beta}$$

$\hat{\alpha}_i$'s are linear

$$\begin{bmatrix} \hat{\mu} \\ \hat{\alpha} \end{bmatrix} = \hat{\beta} = (X^T X)^{-1} X^T y$$

$$= \beta + (X^T X)^{-1} X^T \epsilon$$

$\hat{\beta}$ is multivariate Normal

$$\hat{\alpha}_a - \hat{\alpha}_b = [0 \dots 0 \underset{a^{th}}{1} 0 \dots 0 \underset{b^{th}}{-1} 0 \dots 0] \hat{\beta}$$

$$\sim N(\alpha_a - \alpha_b, \dots)$$

I independent parameters
(N-I) degree of freedom

$$N = \sum_I n_i$$

To compare level 1 with level b ($b \neq 1$) we can use the option contrast μ treatment. This is equivalent to $\binom{k}{2}$ of ways to resolve non identifiability. The R code extracts, gives estimate of μ , $\alpha_b - \alpha_1$, along with standard errors, t -statistic and p -value.

So we can compare level 1 with all other levels. After that we can shuffle (permute) the rows of the data to make a different level as 1st and carry out the same analysis. OR, we can drop some points corresponding to the 1st level & repeat the analysis.

Multiple Tests

There are $\binom{I}{2}$ pairwise tests performed after the ANOVA.

Each test is conducted at significance level α , say 5%.

Even if $\alpha_a = \alpha_b$ (that is, H_a is true), we could still reject it 5% of the time.

If we are doing 10 tests (i.e., $\binom{5}{2}$), the probability of making at least one mistake, even if all null hypotheses were true, is much higher than 5% it could be as high as 50%. Problem with multiple hypothesis testing.

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In March 2012, Amgen reported their attempt to replicate experiments from dozens of biomedical publications.

They examined 53 ‘landmark’ cancer-related preclinical studies, where a drug molecule was tested in animals.

Many of these studies had been published, cited, and followed up — yet, only 6% (11%) of the 53 results could be reproduced.

Conclusion: Drug was not useful and was therefore rejected.

If $k \gg 1$ test are performs each at level α then when all the null hypothesis are true, the probability of rejecting at least one of them is $\gg \alpha$.

$$H_{01}, \dots, H_{0k}$$

$$P(\text{Rejecting } H_{0i} \mid H_{01}, \dots, H_{0k} \text{ are true}) = \alpha$$

$$P(\text{rejecting at least one of } H_{0i} \mid \text{all of } H_{0j} \text{ are true}) > \alpha$$

$$A_i = \text{rejecting } H_{0i}$$

$$P(\cup A_i \mid \text{all } H_{0j} \text{ true}) > P(A_1 \mid \text{all } H_{0j} \text{ true}) = P(A_1 \mid H_{01} \text{ true})$$

under all null hypothesis true.

$$P(\cup A_i) = 1 - P(\cap A_i^c) \qquad (\cup A_i)^c = \cap A_i^c$$

$$= 1 - (1 - \alpha)^k \quad [\text{If all } A_i \text{ are independent}] \text{ may not hold always}$$

$$1 - (1 - \alpha)^k \leq k\alpha$$

Solve α .

Do all tests at α .

For the cancer studies indep of A_i may hold.
for heights vs race, it doesn't hold.

$$\beta_{\text{black}} - \beta_{\text{white}}$$

$$\beta_{\text{mexican}} - \beta_{\text{white}}$$

$$\beta_{\text{black}} - \beta_{\text{mexican}}$$

From Bonferroni

$$\alpha_i = \frac{\alpha}{k}$$

$$P(\cup A_i) \leq \sum P(A_i) \quad (\text{Boole's inequality})$$

Each test is done at very low α .

This gives very nice CI

$$Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

Testing $H_0 : \alpha_2 = \alpha_3$

$$t = \frac{\hat{\alpha}_2 - \hat{\alpha}_3}{se(\hat{\alpha}_2 - \hat{\alpha}_3)} \sim t_d \text{ at level } \alpha/10$$

s^2 estimates σ^2 with d degrees of freedom

$$\binom{5}{2} = 10 \text{ tests.}$$

$$\hat{\alpha}_2 \quad s.e.(\hat{\alpha}_2) \quad cov(\hat{\alpha}_2, \hat{\alpha}_3)$$

$$\hat{\alpha}_3 \quad s.e.(\hat{\alpha}_3)$$

Bonferroni is conservative

$$\sum P(A_i) < \alpha/K \quad \forall i=1,2,\dots,k$$

$$P(A_i) < \alpha \quad i = 1, \dots, K$$

$$\text{Tukey } P\left(\max_{i \neq j} \frac{|\hat{\alpha}_i - \hat{\alpha}_j|}{se} > a\right) = \alpha$$

$a \rightarrow q_\alpha$, as quantile to reject.

Max t -statistic should exceed the cutoff with prob α

$$X_1, \dots, X_k \stackrel{i.i.d}{\sim} N(\mu, \sigma^2)$$

$$T = \frac{\max X_i - \min X_i}{s.e(\max - \min)} \rightarrow \text{Tukey}$$

This is not a t -dist anymore.

quantiles of this "Tukey's dist" are available

HSD (Honest Significant Difference)

Higher cut off than Bonferroni Rejection region (less than α/K area, more than $\frac{\alpha}{K}$ area)

For Bonferroni at level α/K

all K -tests are observed done with Tukey we allow p -values (of original t -distribution) $< \alpha/K$ and some to have p -values $> \alpha/K$

as long as the max satisfies Tukey's condition.

Family wise CI

All H_{oi} are true and rejects at least one erroneously.

Family wise error rate (FWER)

e.g. Bonferroni, Tukey.

This FWER approach is conservative as we are not ready to get even one error.

If K is very large, then the C.I are very wide.

An alternative is to use False Discover Rate (FDR)

The properties of H_{0i} : that are rejected when all H_{0i} are true FDR.

Signal, $\beta_1 - \beta_2 > 0$.

Benjamini-Hochberg procedure

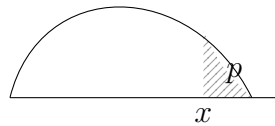
K many test statistic find p-values. $p_i, i = 1, \dots, K$.

Sort them $p_{(1)} < \dots < p_{(K)}$

Reject the corresponding null hypothesis if $p_{(i)} \leq \frac{i}{K}\alpha$ distribution of P-values is uniform.

K-tests

- ① No adjustment, (each is performed at α level might lead to too many rejection.
- ② Bonferroni correction. Perform each test at level $\frac{\alpha}{k}$. pvalue, statistic. Inverse quantile of statistic.



$p < \alpha$, Reject

(corrected) Bonferroni p-value p_k .

- ③ Tukey $\rightarrow$ max of difference; distribution of that uses same as in reference test.
- ④ FWER when k is too large then use FWER leads to very few rejection. $\frac{\alpha}{k}$ is very small.
- ④ FDR (false discovery Rate)

Benjamin Hochberg

Sort p-value

$$p_{(1)} < p_{(2)} < \dots < p_{(k)}$$

$$p_{(i)} \leq \left\lceil \frac{i\alpha}{k} \right\rceil$$

For this race eq.

i , 'none' $\rightarrow$ 7 rejection

2, 'Tukey' $\rightarrow$

1 'Bonferroni' $\rightarrow$ 6 rejection

FDR: K is very large, most H_o are actually true, most rejection are happening due to statistical procedure.

Physically from the problem you know that level are different $\rightarrow$ FWER.